

Experiment No-: 1

Step 1 : Install java jdk 8

First of all you must install Java JDK 8 on your system. You can just type this command to install java jdk on your system.

```
sudo apt install openjdk-8-jdk
```

To check it's there **cd /usr/lib/jvm**

```
ls
```

```
clear
```

```
sudo nano .bashrc
```

Step 2 : Add this configuration on you bash file

Now just open .bashrc file and paste these commands. Use cntrl+shift+v and cntrl+o to write enter cntrl+x to exit

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export PATH=$PATH:/usr/lib/jvm/java-8-openjdk-amd64/bin
export HADOOP_HOME=~/.hadoop-3.2.3/
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
export HADOOP_STREAMING=$HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-3.2.3.jar
export HADOOP_LOG_DIR=$HADOOP_HOME/logs
export PDSH_RCMD_TYPE=ssh
```

install ssh(ssh — secure shell — protocol used to securely connect to remote server/system — transfers data in encrypted form)

```
sudo apt-get install ssh
```

now go to hadoop.apache.org website download the tar file
(hadoop.apache.org — download tar file of hadoop.)

```
tar -zxvf ~/Downloads/hadoop-3.2.3.tar.gz and press enter
```

(Extract the tar file)

```
cd hadoop-3.2.3/
```

cd etc/hadoop

/

ls

sudo nano hadoop-env.sh

JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

now open hadoop-env.sh sudo nano hadoop-env.sh JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64 (set the path for JAVA_HOME)

cntrl+shift+v and cntrl+o to write enter cntrl+x to exit

Step 3 : Add this file in core-site.xml

sudo nano core-site.xml

Now add this configuration in core-site.xml file.

core-site.xml

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>  </property>
  <property>
    <name>hadoop.proxyuser.dataflair.groups</name> <value>*</value>
  </property>
  <property>
    <name>hadoop.proxyuser.dataflair.hosts</name> <value>*</value>
  </property>
  <property>
    <name>hadoop.proxyuser.server.hosts</name> <value>*</value>
  </property>
  <property>
    <name>hadoop.proxyuser.server.groups</name> <value>*</value>
  </property>
</configuration>
```

Step 3 : Add this file in hdfs-site.xml

Now add this configuration in hdfs-site.xml file.

sudo nano hdfs-site.xml

hdfs-site.xml

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
</configuration>
```

Step 4: Add this file in mapred-site.xml

Now add this configuration in mapred-site.xml file.

```
sudo nano mapred-site.xml
```

mapred-site.xml

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>    <value>yarn</value>
  </property>
  <property>
    <name>mapreduce.application.classpath</name>

    <value>$HADOOP_MAPRED_HOME/share/hadoop/mapreduce/*:$HADOOP_MAPRED_HOME/share/
hadoop/mapreduce/lib/*</value>
  </property>
</configuration>
```

Step 4: Add this file in yarn-site.xml

Now add this configuration in yarn-site.xml file.

```
sudo nano yarn-site.xml
```

yarn-site.xml

```
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
  <property>
    <name>yarn.nodemanager.env-whitelist</name>

    <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_P
REP_END_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_HOME</value>
  </property>
</configuration>
```

```
ssh localhost
```

```
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys
hadoop-3.2.3/bin/hdfs namenode -format
```

format the file system

```
export PDSH_RCMD_TYPE=ssh
```

Step 5 : Start hadoop

To start

```
start-all.sh (Start NameNode daemon and DataNode daemon)
```

```
localhost:9870 codewitharjun@cwa:~$  
hadoop fs -mkdir /user  
codewitharjun@cwa:~$ hadoop fs -mkdir /user/arjun.gautam  
codewitharjun@cwa:~$ touch demo.csv  
codewitharjun@cwa:~$ hadoop fs -put demo.csv /user/arjun.gautam
```

stop-all.sh

This is how you can install hadoop on your ubuntu operating system and start using on your system.

← → ↻ http://localhost:9870/explorer.html#/user/swara Sign in

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities ▾

Browse Directory

/user/swara Go!   

Show 25 entries Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	-rw-f--	hadoop	supergroup	0 B	Jan 21 12:54	1	128 MB	demo.csv	

Showing 1 to 1 of 1 entries Previous 1 Next

Hadoop, 2019.

Output-:

